Data Storage

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Lecture Objectives

- Develop an understanding on data storage basics
- Critically discuss partitioning and partition tables
- Understand deletion and persistence
- Connect storage concepts to forensics

Hard Disks

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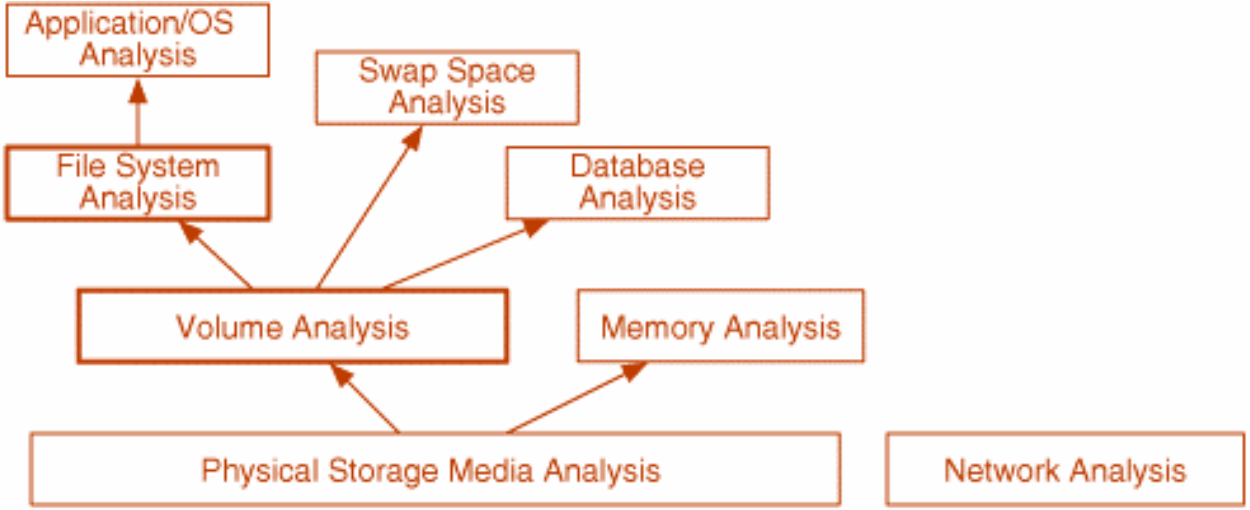
The importance of understanding storage

- Determines what evidence exists
- Explains persistence of deleted data
- Affects recovery possibilities
- Impacts reliability of tools

The need for data analysis

Layers of the analysis based on the design of digital data

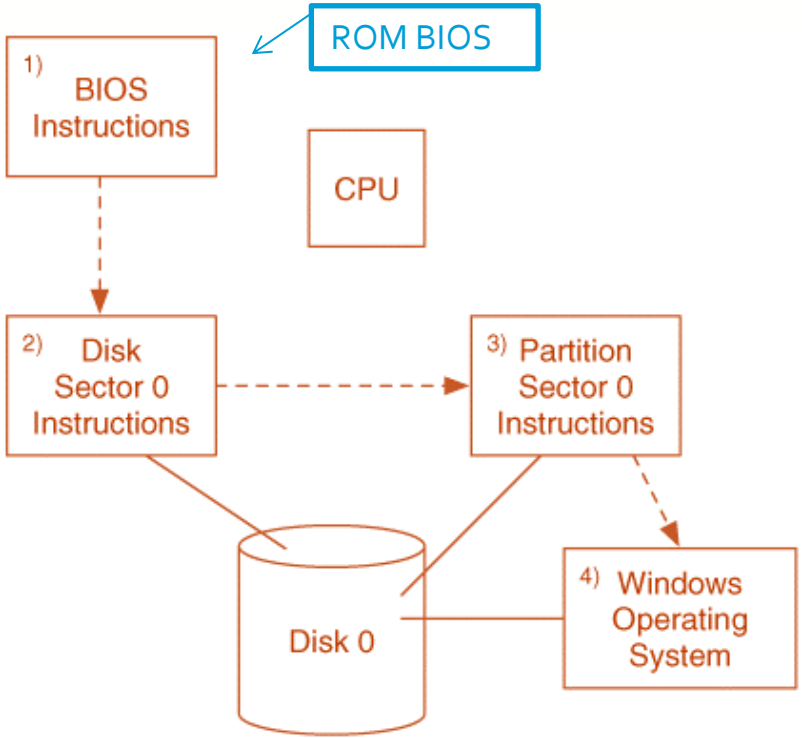


(**File System Forensic Analysis,** Brian Carrier)

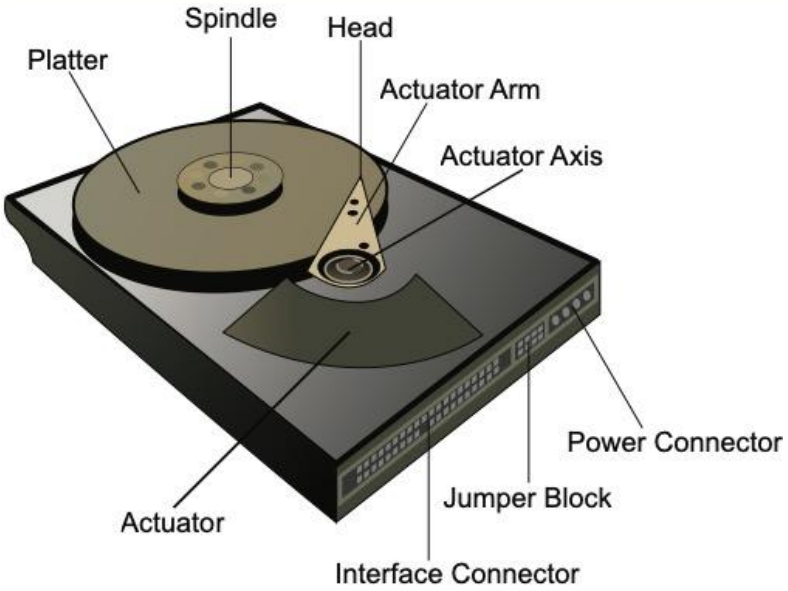
Boot Sequence

POST = Power on Self Test

1. Power supply
2. CPU
3. System Bus (main electrical pathway)
4. System Clock (synch hardware)
5. Video Memory
6. RAM
7. Keyboard
8. Drives
9. CMOS
10. Other System Components

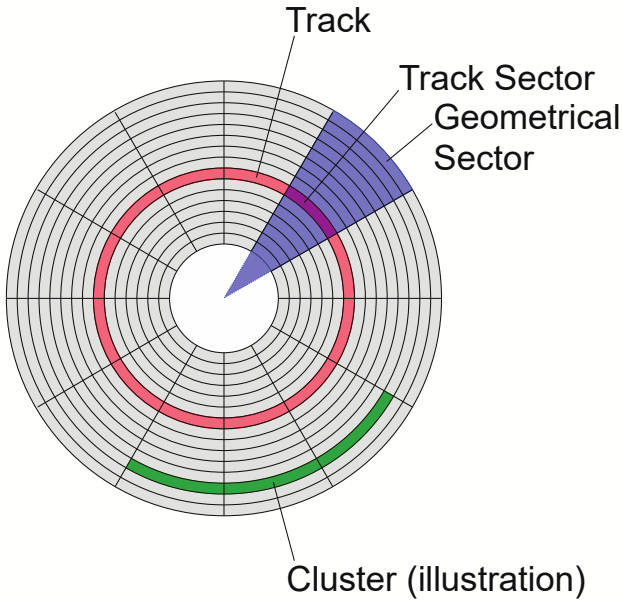


Hard Disk Components



Note that current disks will not have jumper blocks and usually combine power and interface connectors

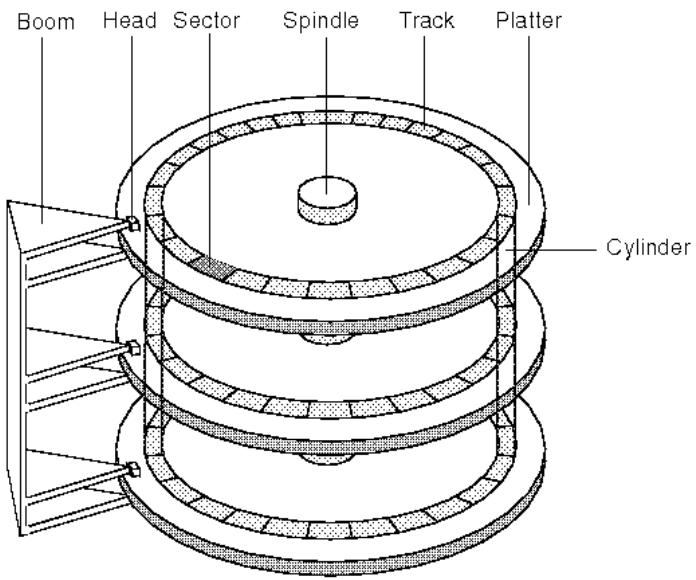
Notional Disk Structure Layout



Note: Modern disks will use non-uniform sector sizes and numbers at different tracks to maximized density

Disk Geometry

- **Tracks:** the recorder information stored on a hard disk. They are concentric disks placed on the surface of each platter.
  - tracks are numbered, starting from zero at the outside of the platter
  - Each track can hold many thousands of bytes of data
- **Sectors:** The smallest unit in a hard disk. Each sector holds 512 bytes of user data
- **Head**: the actual read/write device, located on either side of each platter, operates on the side of the platter it faces
- **Cylinder:** the collection of track numbers located at all of the heads of the disk



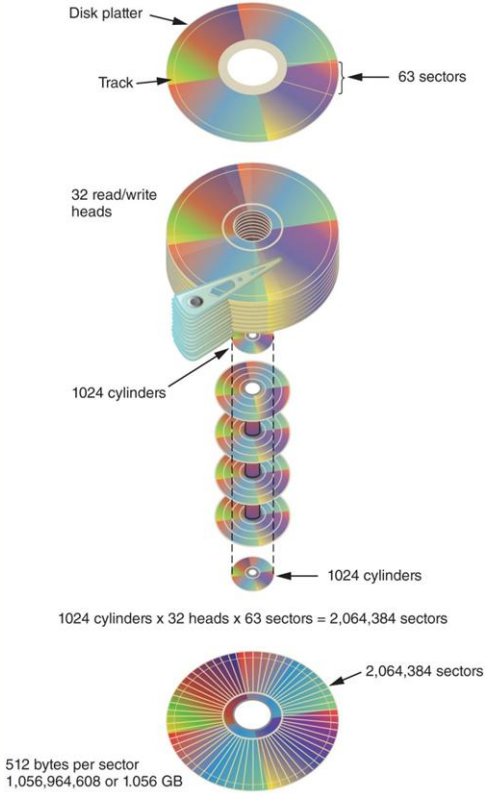
http://docsrv.sco.com:507/en/HANDBOOK/graphics/harddisk.gif

CHS and LBA

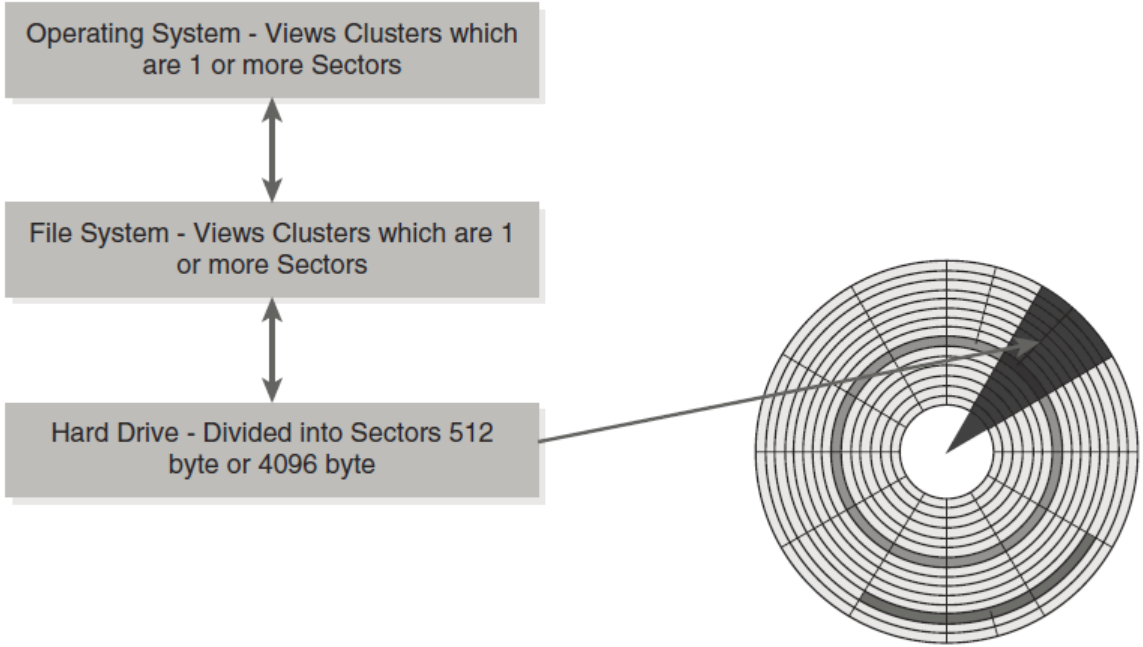
- CHS (Cylinder, Head, Sector)
  - type of sector addressing, no longer used
  - Sectors have an address, starting from 1 at the beginning of each track
  - A sector can be addressed by the (C) for the track, the (H) number for the platter and the (S) to find the sector in the track.
  - Can still be proved helpful in a forensic investigation.
  - Limit 8.1GB disks

- LBA (Logical Block Address)
  - assigns sequential address to each sector
  - the software does not need to know anything about the
  - geometry; it needs to know only a single number (sector address)
  - May not be related to the physical location
  - some file systems and other data structures nowadays that still use the CHS scheme, therefore it may be necessary to convert a CHS address to a LBA address using the following formula

LBA = (((Cylinder * heads_per_cylinder) + Head) * Sectors_per_track) + Sector -1



File Systems and Hard Drives



Solid State Drives

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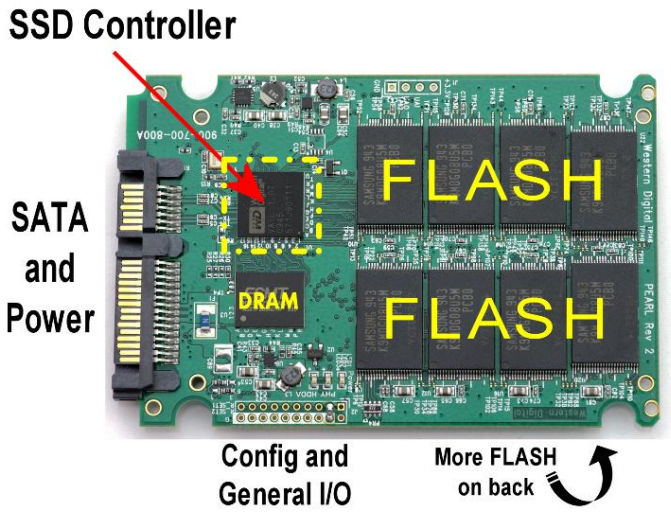
Solid-State Storage Devices

- All flash memory devices have a feature called **wear-leveling**
- An internal firmware feature used in solid-state drives that ensures even wear of read/writes for all memory cells
- When dealing with solid-state devices, making a full forensic copy as soon as possible is crucial
- In case you need to recover data from unallocated disk space

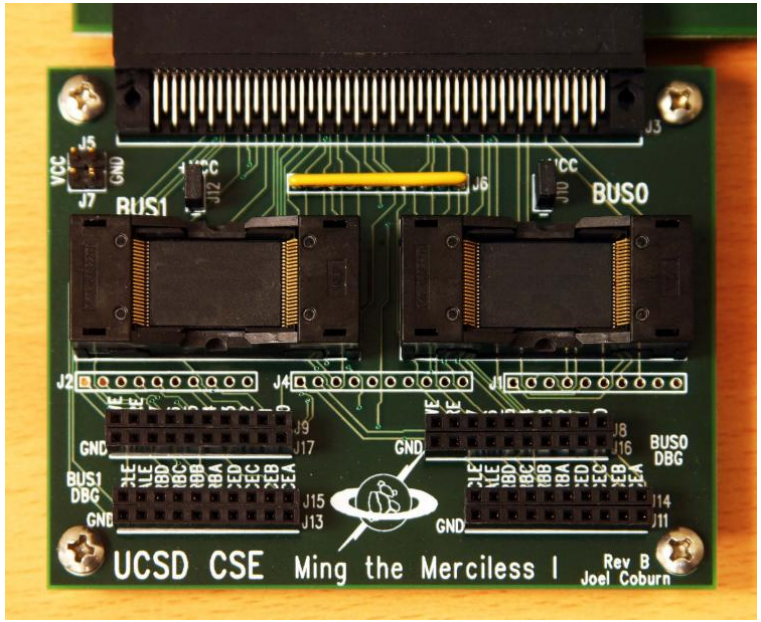
Generic Solid-State Disk (1/2)

SSD components (here: SATA interface)



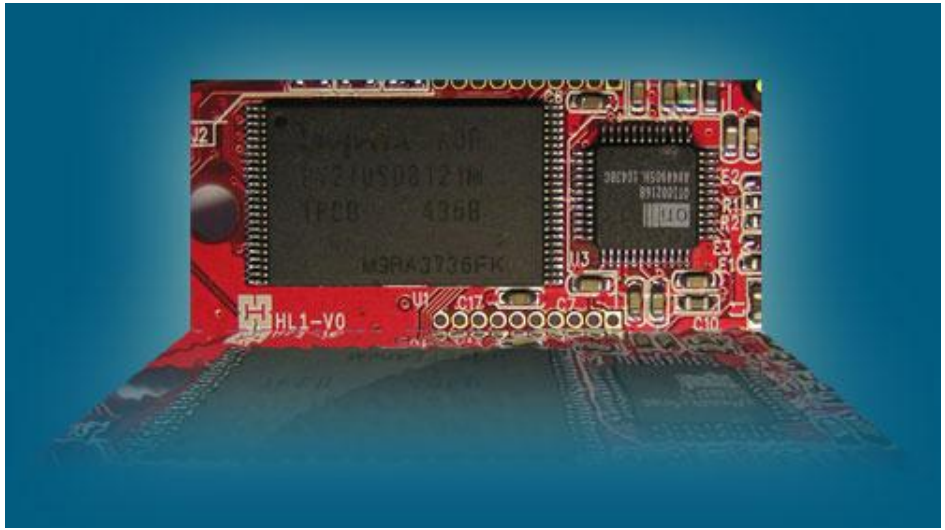
Controller provides external interface support, but particularly FTL adaptation and possibly logic for controlling flash memory

Generic Solid-State Disk (2/2)



Bypassing the FTL requires disconnecting the controller and "manually" accessing flash memory chips (M. Wei et al.: *Reliably Erasing Data From Flash-Based Solid State Drives*, Proc. USENIX FAST 2011)

Typical Managed NAND Flash Memory



The (NAND) Flash memory circuit can be seen on the left; the right-hand circuit provides the FTL adaptation and management of the Flash memory

- The microcontroller may also have further functions, e.g., a USB bus interface

Wear Levelling and Garbage Collection

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Updating data in Flash memory cannot occur directly but consists of an erase-write cycle over a larger (not necessarily the same) block

- After copying and updating, the new block is marked valid (live), the previous instance as invalid (dead) in the spare area
- Once the number of dead pages in a block reaches a threshold, all remaining live pages are re-written elsewhere, and the block is erased so it can be re-written later
- This re-writing occurs under control of the FTL and may occur at different times
- To avoid premature failure, blocks are managed so they are written approximately at the same frequency (wear levelling), this can occur under control of the operating system, or in Flash memory firmware

Forensic Implications of SSD Design

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- Deleted data may be quickly erased
- TRIM can remove data irreversibly
- Imaging captures logical view, not physical reality
- Recovery less reliable than on HDDs

- SSDs prioritise performance, not forensic preservation
- Data persistence cannot be assumed
- Acquisition decisions must be carefully justified

Flash Garbage Collection Implications

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As performance is affected significantly by fragmented block allocation, re-allocation and re-claiming (**garbage collection**) is often performed aggressively

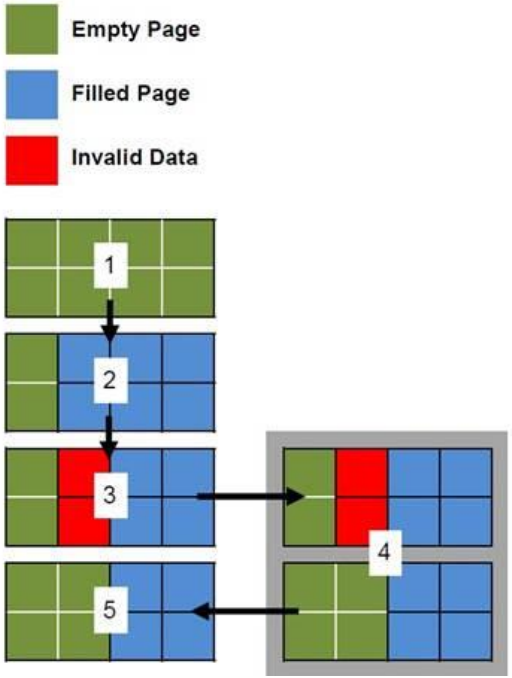
- In case of SSDs or software-controlled FTL ("raw NAND"), this must be deactivated
- For embedded memory ("managed NAND") such as pen drives, low-level recovery is possible when bypassing the management layer

Garbage collection will continue in the background even if no write operations occur as soon as power is applied to a firmware FTL device

- Overwritten instances are then irretrievably lost
- The GC collection mechanism effectively causes **self-corrosion** of court evidence

TRIM Operation

Deletion schematic (similar for FTL-mediated deletion):



1.) SSD pages contain no data

2.) User writes data to SSD pages

3.) User deletes some data. Pages are marked as 'not in use' by the host OS, but data remains on SSD.

4.) TRIM command tells SSD controller that pages contain invalid data. Pages with invalid data are cleaned.

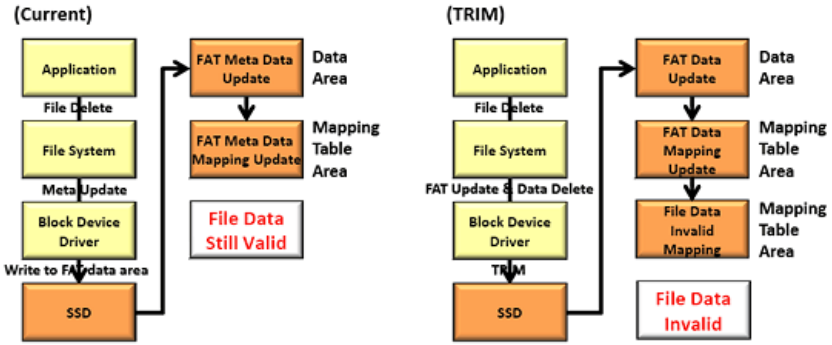
5.) Data is written back to SSD memory cells. The invalid data has been cleaned and data is able to be written to the pages at full speed.

The TRIM Command and Its Consequences

Solid state disks will usually rely on the TRIM command supported by all current-generation operating systems

- The TRIM command informs the SSD that blocks are no longer in use
- The TRIM mechanism duplicates some effects of garbage collection, but causes less wear for the flash cells: The SSD is free to re-use blocks immediately

Use of TRIM indirectly causes changes in how data is (not) retained – consider example of FAT file systems:



Defect Management

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- Similar to magnetic disks, Flash memory must apply **defect management** to ensure continued reliability as devices age
  - **in addition to dynamic error management**
- Blocks may simply be worn out, or have manufacturing defects
  - The latter typically occurs when data in a block has non-recoverable ECC errors
  - Any page producing an error will result in the entire block having to be marked invalid
- Blocks will then be either skipped
  - **Skip Bad Block** or SBB strategy
  - replaced logically from a reserved **Reserve Block Area** (RBA)
- The exact mechanism is vendor-specific, and significantly affects device-internal block allocation
- In either case hidden protected areas or bad-block management areas can be misused

SSD vs HDD: Forensic Comparison

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HDD

- Data stored in fixed physical locations
- Deleted data often persists longer
- More predictable behaviour

SSD

- Data constantly moved internally
- TRIM may erase deleted data quickly
- Recovery is less reliable

Erase, Secure Erase, and Wiping

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- Self-encrypting drives (SEDs) are the most reliable approach
  - Erasing key material ensures data is not readable
- Otherwise, the **Secure Erase** command specified in the ATA standard may not capture all spare capacity, although some manufacturers will ensure all blocks are overwritten
  - Whilst faster than magnetic disks, wiping by overwriting would still be time-consuming
- Purely software-based wiping is bound to be highly ineffective:
  - Not only are write speeds relatively slow, but the FTL activity and remaining spare flash memory would likely leave some memory blocks on the disk (not on the file system) still with data
  - When relying on overwriting – unlike for hard disks – multiple passes may be required if it is not the whole drive being overwritten (probabilistic, up to 20 passes required)

Discussion point

- You are examining a laptop with an SSD. The suspect claims they deleted key files days ago.

Discuss:

- Would you expect recovery to be possible?
- What factors might affect this?
- How would you explain uncertainty in a report?

a collection of consecutive sectors in a volume

Partitions

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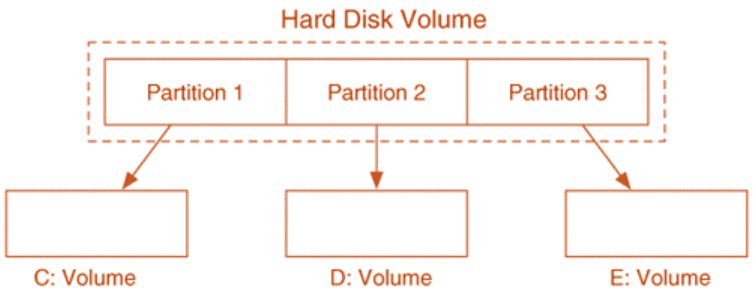
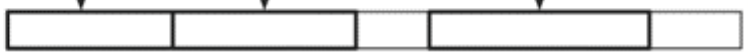
Partitions Basics

- primary and logical partitions
  - organize the layout of a volume
- a partition system is dependent on the operating system and not the type of interface on the hard disk
- when partitions on the disk have been deleted or modified by the suspect, a tool simply cannot locate a partition



Start	End	Type
0	99	FAT
100	249	NTFS
300	599	NTFS

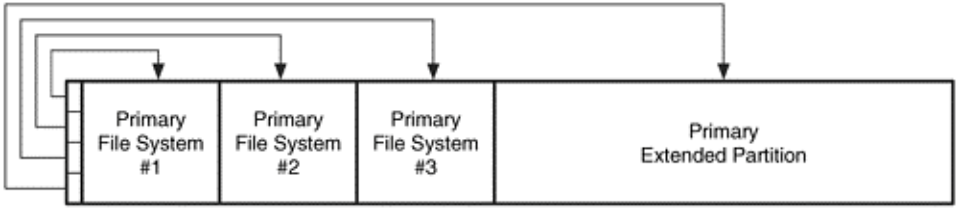


Partition Structure

MBR – Master Boot Record

- the first 512-byte sector (boot code, partition table, signature value)
- sector 0, is where the MBR resides
- There is only one MBR but there can be more than one VBRs

- Primary partition
  - A primary partition contains one file system
  - a hard disk may contain as many as four primary partitions
- Extended partition
  - secondary to the primary partition
  - a hard disk may contain only one, is sub-divided into logical drives, that assign additional drive letters
  - useful if more than 4 primary partitions are needed
  - the first sector of an extended partition holds another partition table



MBR will not recognize or utilise space beyond 2 TB!

Partition Structure

- Recovering deleted partitions
  - Repartitioning, clearing partition structures, corrupted partitions make analysis harder
  - Partition recovering tools
    - Linux tool: `gpart` and `GParted`
- The partition table and boot code require only one sector, yet 63 are typically allocated for both the MBR and extended partitions because the partitions start on a cylinder boundary
  - sector 0 of the extended partition or MBR is used for code and the partition table
  - sectors 1-62 may not be used
  - unused area can be used by additional boot code, but it also may contain data from a previous installation, zeros, or hidden data

Master Boot Record

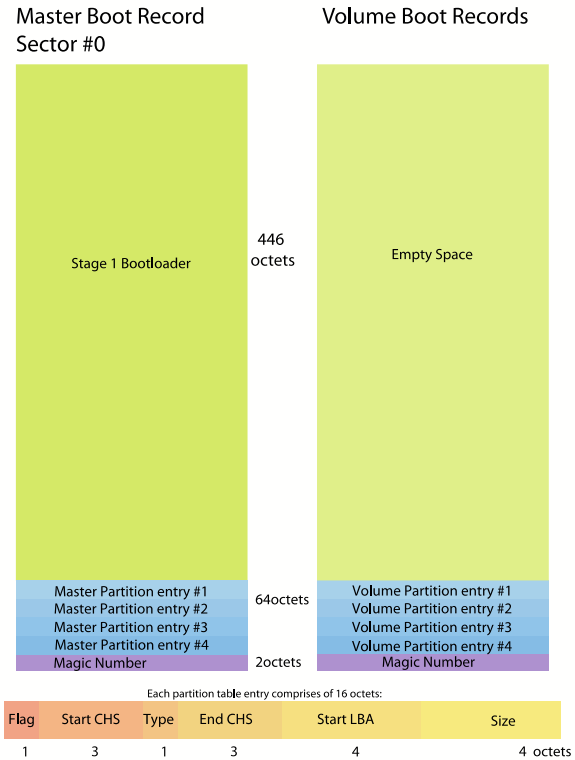
- Special type of boot sector at the very beginning of partitioned secondary storage devices.
  - If you have more than one hard disk, you are going to have more than one MBR
  - Holds information on how the logical partitions and the containing file systems are organized.
  - Operating system-independent chain boot loader that works in conjunction with each partition's Volume Boot Record (VBR) in loading the operating system.
- Boot code:
  - instructions that tell the computer how to process the partition table
  - signature value
- Partition table:
  - 4 entries - each of which can describe a DOS partition
- Signature value
  - Starting CHS address
  - Ending CHS address
  - Starting LBA address
  - Number of sectors in partition
  - Type of partition
  - Flags

MBR Structure

- The first physical sector of the hard disk, sector 0, is where the MBR resides. The MBR is 512 bytes in length.
- The executable code locates and loads the Volume Boot Record from the active partition.
- There is only one MBR but there can be more than one VBRs

Offset	Length	Contents
0	446	Master Boot Record code. This section of code is responsible for locating the partition to boot from and instructing the CPU to continue execution from the start of the File System Boot Sector. More on what this code does is discussed in the next slides. This code contains such errors as "Error Loading Operating System" and "Missing Operating System".
446	16	First Partition Table entry.
462	16	Second Partition Table entry.
478	16	Third Partition Table entry.
494	16	Fourth Partition Table entry.
510	2	55 AA

MBR Organization



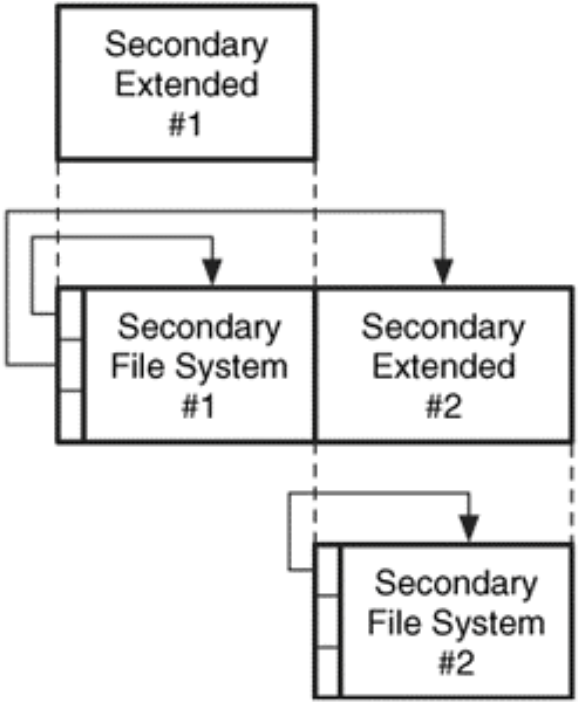
Partition Boot Sector Code

- Code responsible for performing whatever actions are necessary to continue the bootstrap process.
  - different for each operating system.
  - MBR code is OS *independent*,
  - Boot Sector Code is OS *dependent*, *file system* independent.
- The Boot Sector code uses low level BIOS calls
  - locates programs using physical sector information.

Secondary extended and file system partitions

- When an entry for an extended partition is processed, the first sector of the extended partition is read and its partition table entries are processed in the same manner as the MBR.

- When an entry for a non-extended partition is processed, its starting sector and size are displayed. The ending sector address can be determined by adding the starting sector address and the size together and subtracting one.



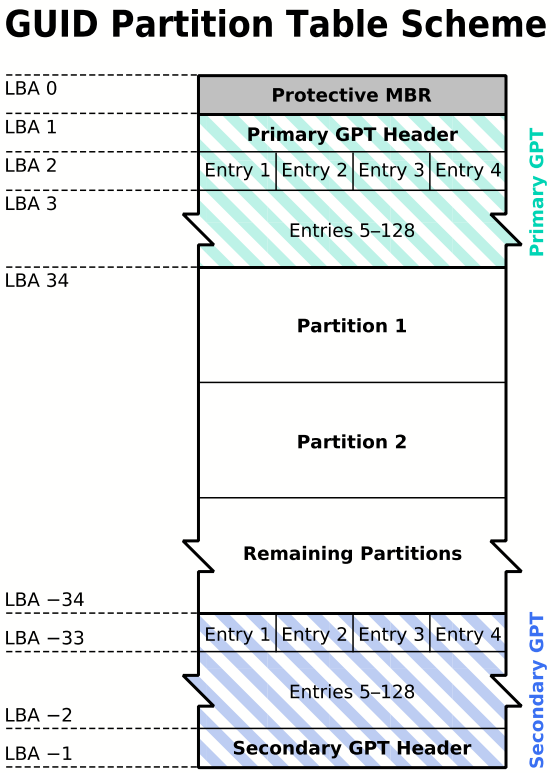
UEFI and GUID Partition Tables

With the introduction of (U)EFI and the GUID partition scheme (**GPT**), size restrictions were eliminated

- GPT always uses LBA (with a "dummy" MBR in sector 0) and has a **GPT Header** in LBA 1 and a **backup** in LBA-1; these identify disk GUID and LBA locations of header and table entries along with number and size of entries
- The GPT Header points to **the Partition Entry Array** (usually in LBA 2)
- Each partition entry contains its type and unique identity as 16-byte GUIDs, the beginning and ending LBA numbers, and an (optional) partition name

The UEFI firmware (i.e., its **Boot Manager** application or layered application within the firmware) can be configured to identify and boot from arbitrary partitions without a need for cascading boot loaders

GPT Record Organization



Unified Extensible Firmware Interface (UEFI)

- UEFI is best thought of as a separate operating system
- Provides both boot and runtime services, and is **extensible** with vendors such as Apple and Microsoft
  - separate boot managers, shells, and user interfaces
- (U)EFI also includes device drivers and graphics interfaces
- Alternatively, to booting from a disk partition, UEFI allows **network booting** via the Pre-Boot Execution Environment (PXE) or network attached storage (iSCSI, FCoE)
  - Windows 8 onwards, recent Linux distributions , x86-based Apple Mac systems
    - UEFI supports a **secure boot** process requiring public key signatures verified against a **platform key** or other firmware-stored key exchange keys
    - although this has been shown to be less than fool-proof

The partition table

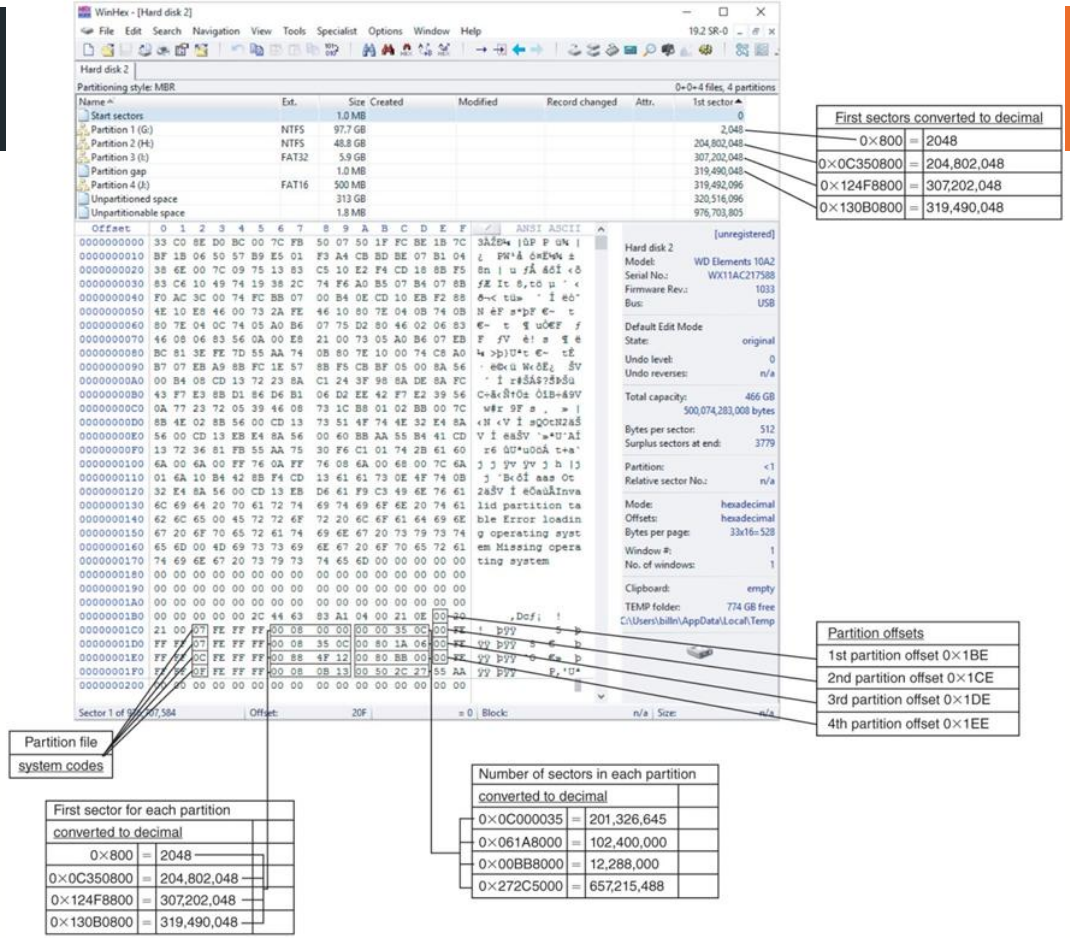


Figure 5-4 The partition table in a hexadecimal editor

Source: X-Ways AG, *www.x-ways.net*

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a storage container that has been formatted with a file system

Volumes

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Volumes

- Digital storage media is organised to allow efficient retrieval of data.
- Volume systems have two central concepts to them:
  1. Assemble multiple storage volumes into one storage volume,
  2. Partition storage volumes into independent partitions.
- The sectors in a volume need not be consecutive on a physical storage device, e.g. hard disk
  - they need to only give the impression that they are.
  - A volume may also be the result of assembling and merging smaller volumes.

Volumes Analysis

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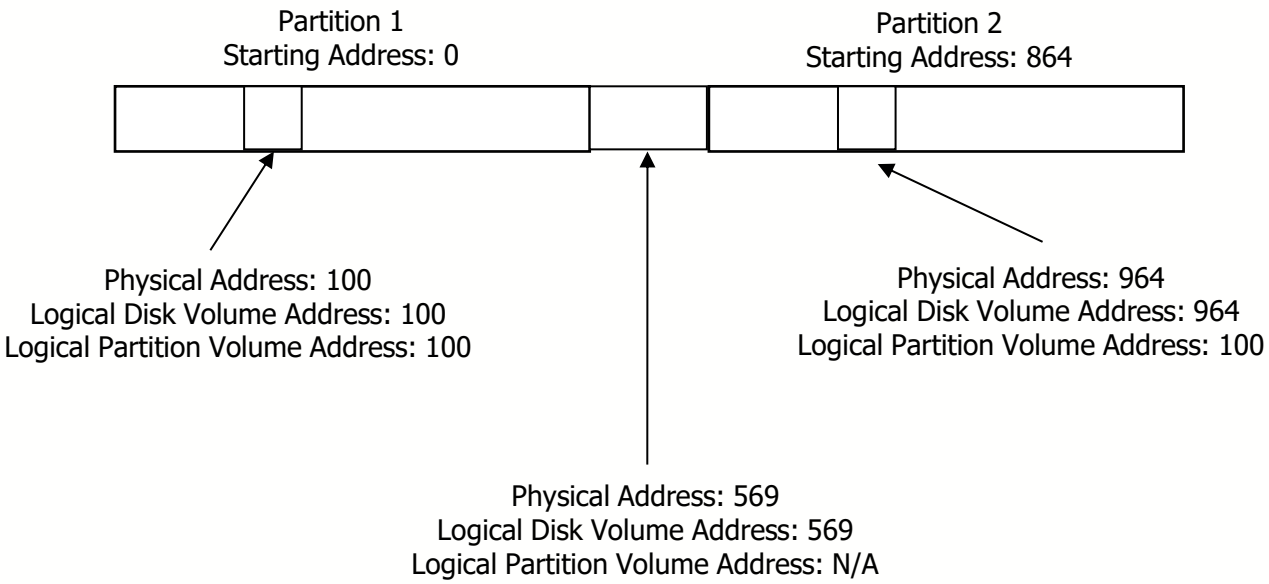
- Volume analysis is looking at the data structures that are involved with partitioning and assembling the bytes in storage devices so that we get volumes.
- During a digital investigation, it is common to acquire an entire disk image and import the image into analysis tools.
- Many digital investigation tools automatically break the disk image into partitions, but sometimes they have problems.
  - What if a disk has become corrupted?
  - What if partitions on the disk have been deleted or modified by the suspect or the tool simply cannot locate a partition?
  - What about the sectors that are not allocated to a partition?

Volume Sector Addressing



Volume Assembly

- Redundancy
  - Data written to more than one disk
  - Raid arrays (for example raid 5)
  - Research different types of raids and their functionality
- Expandability
  - Combine total storage space of multiple volumes to one large “virtual” volume
  - Additional disks can then be added

Low-Level Artefacts

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Low-Level Artefacts (1/2)

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Modern disks contain both a separate operating system and typically also a management interface (e.g., S.M.A.R.T.) which can be interrogated

- These tools and statistics can provide useful evidence in their own right
- More interesting both from forensic and counter-forensic viewpoint: modification of disk firmware
- This can allow access to **bad blocks** or other internal structures that would otherwise be unavailable even to hardware-based imaging tools
- Technically, even these are not necessarily forensic duplicates as not every bit (or sector) from the source disk is available

Different methods apply in case of **hybrid drives** incorporating some solid-state storage – covered in part in discussion of solid-state device forensics

Low-Level Artefacts (2/2)

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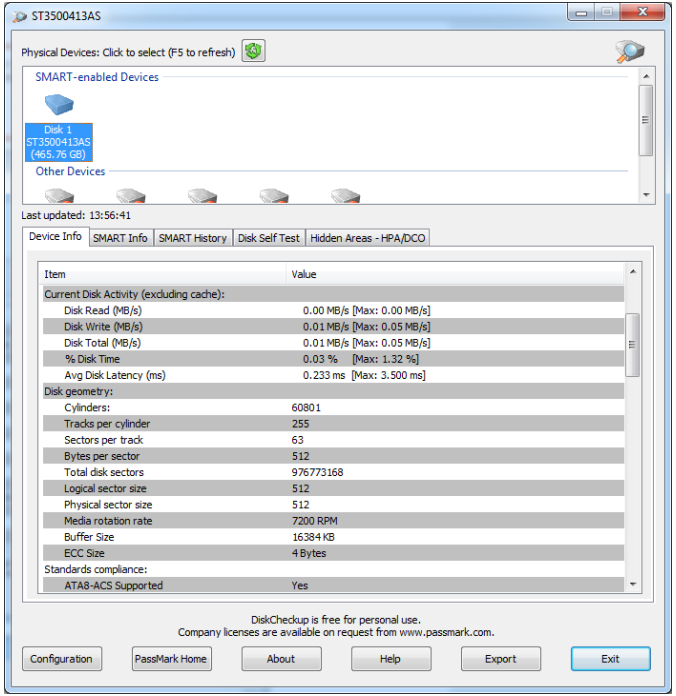
For situations where it is not possible to gain access to the device firmware or if a disk is damaged physically/electronically, further recovery options exist

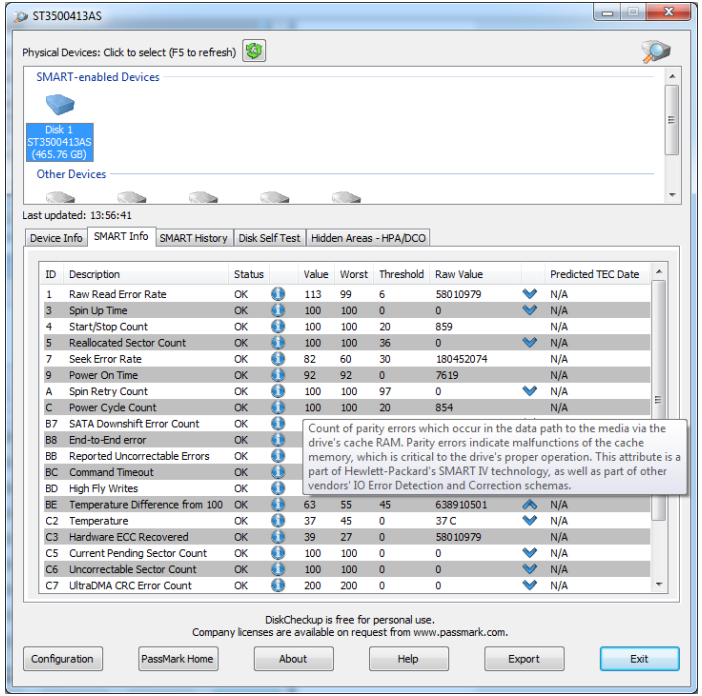
- In case of electronics failure, relevant components can be transplanted from a donor drive
- Defective mechanical components can often also be replaced
- Calibration is increasingly becoming a critical issue as structures become smaller
- Physical destruction, e.g., from head crashes affect larger amounts of data

Depending on the level of physical damage, parts of the media may not be recoverable, or result in substantial errors – somewhat problematic in case of straightforward data recovery, but far more so in case of a legal dispute hinging on the reliability of evidence

Diagnostic Information – SMART (1/2)

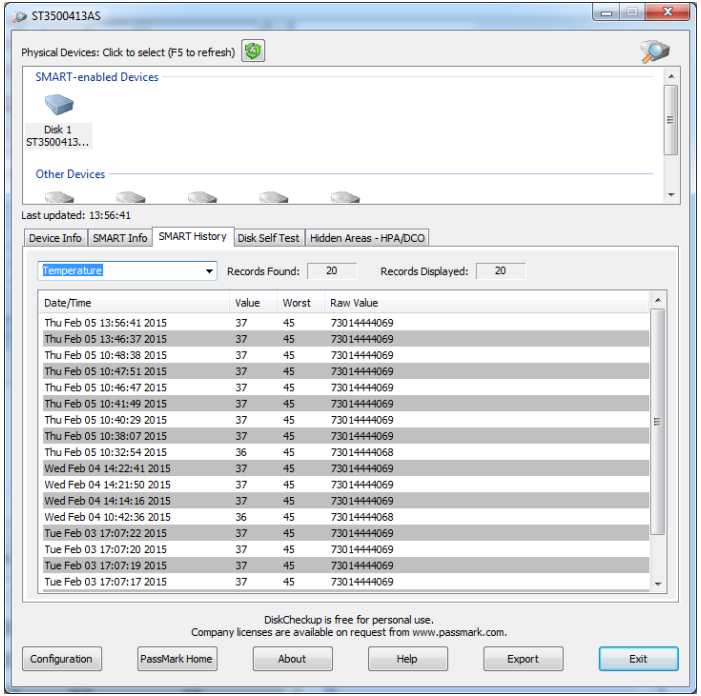
Information readily provided via SMART varies by manufacturer:

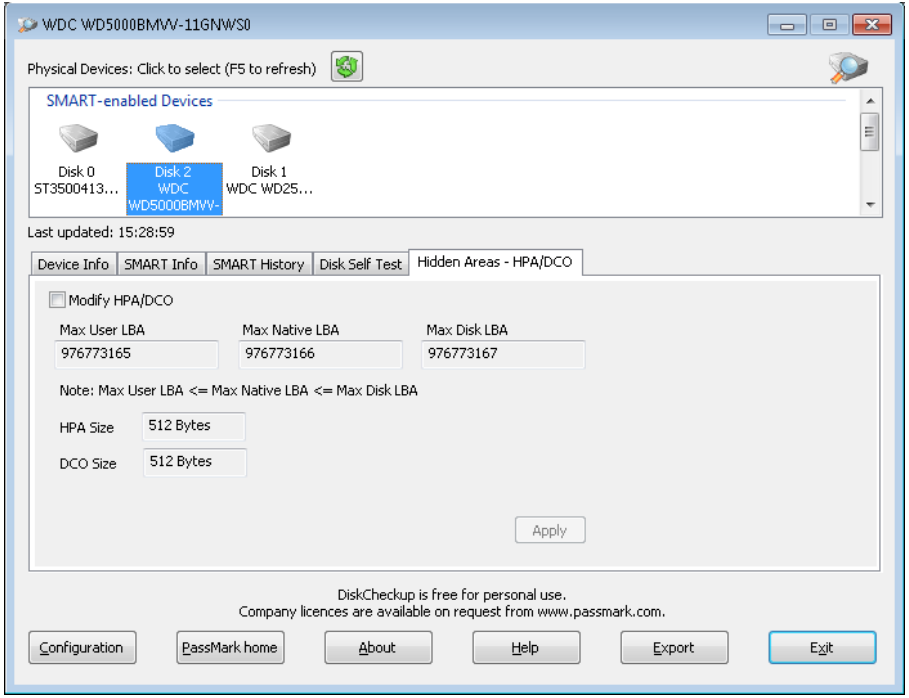




Diagnostic Information – SMART (2/2)

Information readily provided via SMART varies by manufacturer:





Assurance Issues

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Any mechanism providing imaging or **write blocking** must provide assurance of maintaining the above objectives

- Manufacturers may need to provide expert testimony when challenged
- Forensics laboratories may provide test results
- The NIST CFTT (Computer Forensics Tool Testing Program) of the US National Institute of Standards and Technology provides detailed test plans for imaging and write blocking devices
- Hardware-based systems are simple to implement and particularly to validate

Software-based tools must not only be shown to be correct, but the **entire system on which they are running must be validated** – this is rather complicated for general-purpose computers

Duplication and Reconstruction of Volumes (1/2)

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Volume systems may be used to combine multiple sub-volumes into a single volume

- All components required for duplication must be identified and recorded clearly
- In case of aggregate volumes, the **configuration** of components (hardware and software) used needs to be recorded
- Complex systems such as **storage area networks** are extremely difficult to reconstruct otherwise as even the topology can be modified dynamically
- Identification of storage locations for logical and hardware volume and disk management may need to be first stage in forensic analysis
- Once this has occurred, file system types can be identified (often heuristically)

Duplication and Reconstruction of Volumes (2/2)

As in the case of files, disks and partitions may contain un-allocated as well as **slack space**

- These can be used by arbitrary file systems (e.g., implemented directly by application code not physically present on the target system) and must be analyzed separately

Other counter-forensic techniques may rely on allocation strategies pursued by the regular operating system

- Overwriting space in a volume that is known or assumed to be unlikely to be overwritten (e.g., on large volumes or in case of modest fragmentation)
- These techniques also must bypass (or at least not interfere with) existing file systems

Summary

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Summary

- Hard drives and storage architecture
- Single drive volume management
- Insights on the SSD architecture
- MBR and GPT structure

Questions?

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